SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code	**CSA4315**	**Course Title**	**Internet Programming**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 3: Application Based Questions**
Weightage	**40% of CIA**	**Total Marks**	**100**
CO1 Statement	*Explain the fundamentals of web architecture, client-server communication, Internet protocols, HTTP request and response messages, and configure LAMP/WAMP environments for web application development and deployment.* **(BL3)**		
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Code of Conduct:

I,L KARTHICK BALAJI, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

KARTHICK

Question 14

A company wants to provide website hosting, email services, and file-transfer services. Identify the suitable protocols for each service and justify your choices.

1. Introduction

Every Internet service works on the **client–server model**: a client (a browser, mail application or FTP client) sends a request and a server responds with the requested service. Communication between them follows the **TCP/IP protocol suite**, a layered set of rules in which application-layer protocols such as HTTP, SMTP and SFTP define how each individual service behaves, while the transport layer protocol **TCP** provides reliable, ordered and error-checked delivery underneath them.

A company that wants to offer **website hosting**, **email services** and **file-transfer services** must therefore select, for each service, the protocol that matches its purpose, its security needs and the clients that will use it. This answer identifies the suitable protocols, explains how each works, justifies every choice, and describes how the services can be deployed on a LAMP/WAMP-based server environment.

2. Web Architecture and Client–Server Communication

A typical Internet service is organised in three tiers, each with a clear responsibility:

- **Client tier (presentation):** the user's browser, mail application or FTP/SFTP client. It builds the request, sends it over the network and displays the result.
- **Server tier (application/logic):** the web server (Apache or Nginx), the mail server (Postfix and Dovecot) and the file server (OpenSSH or vsftpd). It receives requests, applies the application logic and produces responses.
- **Data tier (storage):** the MySQL database, the mailboxes and the file storage that hold website data, user messages and uploaded files.

Request flow (what happens when a user opens a website)

- **Step 1 – DNS lookup:** the browser asks the DNS server to convert the domain name (www.company.com) into an IP address.
- **Step 2 – TCP connection:** the client opens a connection to the server using the TCP three-way handshake (SYN, SYN-ACK, ACK) on port 80 or 443.
- **Step 3 – TLS handshake (HTTPS only):** the server presents its digital certificate, the client verifies it, and both agree on session keys to encrypt the traffic.
- **Step 4 – Request:** the browser sends an HTTP request message for the required page.
- **Step 5 – Processing:** the web server runs the script (for example PHP) and queries the database if required.
- **Step 6 – Response:** the server returns an HTTP response with a status code and the page content, which the browser renders.

Where the protocols sit in the TCP/IP model

Layer	Examples	Role in the company's services
Application	HTTP, HTTPS, SMTP, IMAP, POP3, FTP, SFTP, DNS	Define the rules of the web, email and file-transfer services
Transport	TCP (and UDP for DNS queries)	Reliable, ordered delivery; port numbers identify each service
Internet	IP, ICMP	Addressing and routing of packets between client and server
Network access	Ethernet, Wi-Fi	Physical transmission of frames over the local network

3. Summary of Suitable Protocols

Service	Protocol(s)	Port / Transport	Purpose
Website hosting	HTTP, HTTPS (HTTP over TLS)	80 / 443 – TCP	Delivers web pages and web applications to browsers securely
Email services	SMTP; IMAP and/or POP3 (with SSL/TLS)	25, 587, 465 (SMTP); 143, 993 (IMAP); 110, 995 (POP3) – TCP	SMTP sends and relays mail; IMAP/POP3 let users retrieve mail
File-transfer services	SFTP (preferred), FTPS, FTP	22 (SFTP); 21 and 990 (FTP/FTPS) – TCP	Uploading and downloading files between client and server
Supporting service	DNS	53 – UDP/TCP	Maps domain names and mail (MX) records to IP addresses

4. Website Hosting – HTTP / HTTPS

4.1 How HTTP works

HTTP (HyperText Transfer Protocol) is the application-layer protocol of the World Wide Web. It is a **stateless**, text-based **request–response** protocol: the client sends a request message made of a request line, headers and an optional body, and the server answers with a response message made of a status line, headers and a body. Because it is stateless, the server does not remember earlier requests; sessions and cookies are used when the website needs to remember a user.

HTTP request and response messages

```
REQUEST (client -> server)
GET /index.html HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Accept: text/html
Connection: keep-alive

RESPONSE (server -> client)
HTTP/1.1 200 OK
Date: Thu, 22 Oct 2026 10:30:00 GMT
Server: Apache/2.4
Content-Type: text/html; charset=UTF-8
Content-Length: 1250

<html> ... page content ... </html>
```

HTTP methods

Method	Use
GET	Retrieve a page or resource; parameters are sent in the URL
POST	Submit data (for example a login or registration form) in the request body
PUT / PATCH	Create or update a resource on the server
DELETE	Remove a resource
HEAD	Retrieve only the response headers, without the body

HTTP status codes

Code	Meaning
1xx	Informational – the request was received and is being processed
200 OK	The request succeeded and the content is returned
301 / 302	The resource has moved permanently / temporarily (redirection)
403 / 404	Access forbidden / page not found (client errors, 4xx)

Code	Meaning
500 / 503	Internal server error / service unavailable (server errors, 5xx)

4.2 HTTPS – secure HTTP

HTTPS is HTTP carried inside a **TLS/SSL** encrypted channel on port 443. During the TLS handshake the server proves its identity with a certificate issued by a trusted Certificate Authority, and both sides create session keys. After that, all HTTP messages are encrypted, so passwords, payment details and personal data cannot be read or altered by anyone on the network.

4.3 Justification for choosing HTTP/HTTPS

- HTTP is the **only standard protocol** understood by every browser and web server, so the website is reachable from any device.
- It is **simple, stateless and scalable**: many requests can be handled by load-balanced servers, and responses can be cached to improve speed.
- It supports **cookies, sessions and dynamic content**, which are needed for logins, shopping carts and database-driven pages.
- **HTTPS** gives confidentiality, integrity and server authentication. It protects users, builds trust, is required by modern browsers for many features, and improves search-engine ranking.
- Plain HTTP should be kept only to redirect visitors automatically to HTTPS.

4.4 Deployment on a LAMP / WAMP environment

LAMP stands for **L**inux, **A**pache, **M**ySQL and **P**HP and is used for production hosting. **WAMP** replaces Linux with **W**indows and is mainly used for development and testing. The steps to host the website are:

- Install Apache, MySQL and PHP (on Linux with the package manager, or on Windows using the WAMP package) and start the services.
- Copy the website files into the document root (for example /var/www/html, or the www folder in WAMP).
- Create a **virtual host** so that the domain name points to the website folder:

```
<VirtualHost *:443>
    ServerName www.company.com
    DocumentRoot /var/www/company
    SSLEngine on
    SSLCertificateFile    /etc/ssl/certs/company.crt
    SSLCertificateKeyFile /etc/ssl/private/company.key
</VirtualHost>
```

- Redirect port 80 to port 443, and allow ports 80 and 443 in the firewall.
- Create the database and a restricted MySQL user for the website, then add a DNS **A record** that points the domain to the server's IP address.

5. Email Services – SMTP, IMAP and POP3

5.1 Components of an email system

- **MUA (Mail User Agent):** the user's mail application, such as Outlook or Thunderbird.
- **MSA/MTA (Mail Submission/Transfer Agent):** the server software (Postfix) that accepts mail from the user and forwards it towards the destination.
- **MDA (Mail Delivery Agent):** the software (Dovecot) that places the message in the recipient's mailbox.

Mail flow

Sender's mail client → **SMTP** → sender's mail server → DNS **MX** lookup → **SMTP** → recipient's mail server → mailbox → the recipient retrieves the message using **IMAP** or **POP3**.

5.2 Protocols used

- **SMTP (Simple Mail Transfer Protocol):** a **push** protocol used to send mail from the client to the server and to relay it between servers. Authenticated submission uses port 587 (STARTTLS) or 465 (SSL/TLS); port 25 is used for server-to-server delivery.
- **IMAP (Internet Message Access Protocol):** a **pull** protocol that keeps messages on the server, synchronises folders and read/unread status across devices, and supports server-side search.
- **POP3 (Post Office Protocol v3):** downloads messages to one device and normally deletes them from the server, which suits offline use and limited server storage.

Feature	IMAP (143 / 993)	POP3 (110 / 995)
Mail storage	Kept on the server	Downloaded to the device
Multi-device access	Fully supported and synchronised	Not suitable
Server storage needed	High	Low
Offline reading	Only for cached messages	Full, after download
Best for	Modern users with phone, laptop and web	Single device, offline reading

5.3 Justification for choosing SMTP, IMAP and POP3

- Email needs separate protocols for **sending** (SMTP) and **receiving** (IMAP/POP3); SMTP alone cannot deliver mail to a user's inbox on demand, and IMAP/POP3 cannot send mail.
- **IMAP is recommended as the default** because users now read mail on several devices; **POP3** is offered as an option for users who want mail stored locally.
- All three protocols must run over **SSL/TLS** (ports 465/587, 993 and 995) so that passwords and message content are encrypted.
- All are open, standard protocols supported by every mail client, which guarantees interoperability with other mail servers on the Internet.

5.4 Deployment and mail security

- Install **Postfix** (SMTP) and **Dovecot** (IMAP/POP3), and create user mailboxes.
- Publish **MX records** in DNS so that other servers know where to deliver mail for the company's domain.
- Add **SPF, DKIM and DMARC** records to verify the sender, sign outgoing mail and reduce spam and spoofing.
- Require authentication for sending, disable open relay, and install spam and virus filtering.

6. File-Transfer Services – SFTP / FTPS / FTP

6.1 Protocols

- **FTP (File Transfer Protocol):** designed for efficient transfer of large files. It uses a **control connection** (port 21) for commands and a separate **data connection** for the files, in active or passive mode. Usernames, passwords and data are sent in **plain text**, so FTP alone is insecure.
- **FTPS (FTP over SSL/TLS):** FTP with TLS encryption added. It keeps all FTP features, but needs extra firewall ports for the passive data connections.
- **SFTP (SSH File Transfer Protocol):** a file-transfer protocol that runs inside an **SSH** session on a single port (22). It provides strong encryption, password or key-based authentication, and data-integrity checks.

Feature	FTP	FTPS	SFTP
Encryption	None	SSL/TLS	SSH
Ports	21 + data port	21/990 + data ports	22 only
Authentication	Password (clear text)	Password / certificate	Password / SSH key
Firewall friendly	No	Moderate	Yes

Feature	FTP	FTPS	SFTP
Suitability	Public, non-sensitive files	Legacy FTP systems	Confidential business data

6.2 Justification for choosing SFTP (with FTPS/FTP as options)

- File transfer often carries **confidential business data**, so a secure protocol is essential. **SFTP is chosen as the main protocol** because everything, including login credentials, is encrypted.
- Using a **single port (22)** makes it easy to configure firewalls and reduces the attack surface.
- **FTPS** is kept as an alternative for partners who still use FTP clients, and **plain FTP** is limited to anonymous, public downloads that contain no sensitive data.
- FTP-family protocols are reliable for large files because they run over TCP and can resume interrupted transfers.

6.3 Deployment

- Install the **OpenSSH server** for SFTP (or **vsftpd** for FTP/FTPS) and create separate user accounts.
- Use a **chroot jail** so each user can see only their own folder:

```
Match Group sftpusers
    ChrootDirectory /srv/sftp/%u
    ForceCommand internal-sftp
    AllowTcpForwarding no
```

- Enable key-based login, disable anonymous access, set upload quotas and keep logs of all transfers.

7. Comparison of Services and Firewall Ports

Service	Main protocol	Server software	Ports to open
Website hosting	HTTPS (HTTP redirect)	Apache + MySQL + PHP	80, 443
Email sending	SMTP over TLS	Postfix	25, 587, 465
Email retrieval	IMAP / POP3 over TLS	Dovecot	993, 995 (143, 110 optional)
File transfer	SFTP (FTPS optional)	OpenSSH / vsftpd	22 (21, 990 optional)
Name resolution	DNS	BIND / provider DNS	53

8. Why Other Protocols Are Not Suitable

- **Telnet, plain FTP and unencrypted HTTP/POP3/IMAP:** send credentials and data in clear text and can be intercepted.
- **TFTP:** has no authentication and runs over UDP, so it is unreliable and insecure for business file transfer.
- **UDP-based transport for these services:** does not guarantee delivery or ordering, which web pages, mail and files require.
- **POP3 alone for modern users:** does not synchronise mail across devices, so IMAP is preferred.

9. Security, Performance and Maintenance Recommendations

- **Encryption:** use TLS certificates for HTTPS, SMTP/IMAP/POP3 and FTPS, and SSH keys for SFTP.
- **Firewall:** open only the required ports and block everything else; keep the operating system and server software patched.
- **Authentication and access control:** use strong passwords or keys, multi-factor authentication for administrators, and role-based permissions.
- **Performance and scalability:** enable HTTP caching and compression, use a load balancer or CDN for the website, and set mailbox and upload quotas.
- **Reliability:** take regular backups of the website, database, mailboxes and files, and monitor server logs for

suspicious activity.

10. Conclusion

The company should use **HTTPS** for website hosting, **SMTP with IMAP/POP3 (secured by TLS)** for email, and **SFTP (or FTPS)** for file transfer, supported by **DNS** for name resolution. The website can be deployed on a **LAMP/WAMP** environment, while the mail and file servers run on the same or separate Linux servers. These standard protocols are reliable (TCP), secure (TLS/SSH), widely supported and scalable, and each one is the best match for the requirements of its service, which justifies the choices made.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Web Architecture & Client–Server Model	**20%**	Explains client–server and 3-tier architecture and the request flow for all three services with clear examples.	Explains client–server model and request flow for all services with minor gaps.	Describes client–server model only partly; flow for some services is unclear.	Little or no explanation of web or client–server architecture.	**/20**
Internet Protocol Selection	**20%**	Correct protocols for all three services (HTTPS, SMTP/IMAP/POP3, SFTP/FTPS) with ports and TLS/SSH security.	Correct protocols for all services; ports or secure variants partly missing.	Correct protocols for only one or two services.	Protocols missing or incorrect.	**/20**
HTTP Request & Response Messages	**20%**	Shows and explains complete HTTP request and response with method, headers, status codes and body.	Explains HTTP request/response with methods and status codes; example is basic.	Mentions HTTP request/response without proper message structure.	HTTP messages not explained.	**/20**
LAMP / WAMP Configuration & Deployment	**20%**	Describes LAMP/WAMP components and configuration steps to deploy web, mail and file-transfer services.	Describes LAMP/WAMP components and basic deployment of the website.	Names LAMP/WAMP components with little configuration detail.	No relevant environment configuration.	**/20**
Justification, Security & Presentation	**20%**	Every choice justified with security, reliability and scalability; well organised and error-free.	Choices justified with reasons; presentation mostly clear.	Justification is brief or generic; organisation is weak.	No justification; poorly organised.	**/20**